TC789 — TEKNOLOGI BLOCKCHAIN

CHAPTER 3

# Symmetric Cryptography

Foundations of Cryptographic Security for Blockchain

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# Learning Objectives

01

Understand the key cryptography security services

02

Explain stream ciphers vs. block ciphers and their differences

03

Describe block cipher modes of operation (ECB, CBC, CTR)

04

Trace the evolution from DES to 3DES to AES

05

Explain AES internal round operations (4 key steps)

06

Connect symmetric crypto to blockchain use cases

# Chapter Overview — Lecture Flow

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| **Math Foundations** | **Crypto Services** | **Symmetric Ciphers** | **Cipher Modes** | **DES & AES** |
| Sets, groups, fields, modular arithmetic | Confidentiality, integrity, auth, non-repudiation | Stream ciphers vs. block ciphers | ECB, CBC, CTR and others | History, design, AES internals |

# Mathematical Foundations

| | |
|---|---|
| **Set** | A collection of distinct objects, e.g. X = {1, 2, 3} |
| **Group** | A commutative set with one closed, associative operation having identity and inverse elements |
| **Field** | A set with both additive and multiplicative groups — distributive law applies |
| **Finite Field** | A field with a finite number of elements (Galois field) — critical for AES and ECC |
| **Abelian Group** | A group where the operation is commutative: A × B = B × A |
| **Modular Arith.** | "Clock arithmetic" — numbers wrap at a fixed modulus, e.g. 50 mod 11 = 6 |

# What Cryptography Provides

| | |
|---|---|
| C | **Confidentiality**<br>Information accessible only to authorized entities |
| I | **Integrity**<br>Information modifiable only by authorized entities |
| A | **Authentication**<br>Entity authentication & Data origin authentication — who is communicating and is the data genuine? |
| N | **Non-repudiation**<br>An entity cannot deny a previous action — cryptographic evidence is generated |
| Ac | **Accountability**<br>Actions can be traced to the responsible party via detailed audit logs |

# Symmetric Cryptography — Core Concept




- Also known as shared key cryptography or secret key cryptography
- The key must be securely exchanged before any communication begins
- Two cipher types: Stream Ciphers (bit-by-bit) and Block Ciphers (block-by-block)
- Block ciphers: DES, 3DES, AES | Stream ciphers: RC4, A5

# Stream Ciphers

- Encrypt plaintext BIT BY BIT using a keystream
- Encryption = Decryption (XOR / modulo-2 addition)
- Security depends entirely on keystream randomness
- **Two subtypes:**
  - Synchronous — keystream depends only on the key
  - Asynchronous — keystream depends on key + ciphertext
- Common examples: RC4, A5 (2G GSM)
- RC4 was used in WEP — now deprecated/broken

## XOR Operation

| | | | | |
|---|---|---|---|---|
| Plaintext | 1 | 0 | 1 | 1 |
| Keystream | 0 | 1 | 1 | 0 |
| **Ciphertext** | 1 | 1 | 0 | 1 |
| **Decrypt (XOR)** | 1 | 0 | 1 | 1 |

# Block Ciphers

- Encrypt plaintext in fixed-length BLOCKS
- Encrypt block-by-block (not bit-by-bit)
- Built using Feistel Cipher or Substitution-Permutation Network (SPN)
- **Feistel Cipher (Horst Feistel):**
  - Splits data into Left + Right halves
  - Multiple rounds of keyed round functions
  - Confusion via S-boxes — hides the key
  - Diffusion via permutation — spreads plaintext influence
  - Decryption = reverse of encryption process
- Examples: DES (Feistel), AES (SPN)



# Block Cipher Modes of Operation

| | | |
|---|---|---|
| ECB | **Electronic Code Book** | NOT RECOMMENDED |
| | Each block encrypted independently — identical plaintext blocks produce identical ciphertext. Reveals patterns. | |
| CBC | **Cipher Block Chaining** | RECOMMENDED |
| | Each block XOR'd with previous ciphertext before encryption. Uses a random IV. Widely used in practice. | |
| CTR | **Counter Mode** | PARALLELIZABLE |
| | Nonce + counter generates a keystream — converts block cipher into stream cipher. Highly parallelizable. | |
| MAC | **Message Authentication** | INTEGRITY |
| | Produces a cryptographic checksum (MAC) to verify message integrity using the CBC-MAC approach. | |
| Hash | **Cryptographic Hash Mode** | ONE-WAY |
| | Block cipher used as a compression function to produce a fixed-length hash of an arbitrarily long plaintext. | |

# Data Encryption Standard (DES)

| 1977 | 1980s | 1998 | 1999+ | 2001 |
|---|---|---|---|---|
| DES adopted by NIST | Widespread use in industry | EFF "Deep Crack" breaks DES | 3DES: 168-bit key introduced | AES replaces DES globally |

- **DES uses a 56-bit key — only $2^{56}$ possible keys — vulnerable to brute force**
- **July 1998: EFF "Deep Crack" machine cracked DES in under 22.5 hours**
- 3DES: applies DES three times with three 56-bit keys (168-bit total key length)
- 3DES limitation: slow performance and 64-bit block size — not suitable for modern use
- Key lesson: cryptographic security degrades as computing power grows

# Advanced Encryption Standard (AES)

| | |
|---|---|
| **Algorithm** | Rijndael |
| **Inventors** | Joan Daemen & Vincent Rijmen |
| **Standardized** | NIST, 2001 |
| **Block size** | 128 bits (fixed) |
| **Key sizes** | 128 / 192 / 256 bits |
| **Known attacks** | None more effective than brute force |

| Key Size → | Number of Rounds |
|---|---|
| 128-bit | 10 rounds |
| 192-bit | 12 rounds |
| 256-bit | 14 rounds |

## Blockchain Connection

- Bitcoin Core encrypts wallet.dat using AES-256-CBC
- Protects private keys stored locally on the device
- AES used wherever symmetric encryption is needed in blockchain

# How AES Works — Internal Round Operations

## 1 AddRoundKey

XOR the 4×4 state array with a round subkey derived from the master key using the Key Schedule.

## 2 SubBytes

Replace each byte of the state using an S-box lookup table. Achieves CONFUSION by hiding key-ciphertext relationship.

## 3 ShiftRows

Cyclically shift each row of the state to the left by its row index (0,1,2,3 positions). Achieves DIFFUSION.

## 4 MixColumns

Apply a linear transformation column-by-column. Achieves DIFFUSION. (Omitted in the final round.)

**Final round: Steps 1 → 2 → 3 → 1 (MixColumns is SKIPPED and replaced by AddRoundKey)**

# Key Terms & Definitions

### Symmetric Crypto

Encryption and decryption use the same shared secret key

### Stream Cipher

Encrypts plaintext bit-by-bit using a generated keystream (XOR operation)

### Block Cipher

Encrypts plaintext in fixed-size blocks using Feistel or SPN design

### Confusion

Hides key-ciphertext relationship — achieved by S-box substitution in SubBytes

### Diffusion

Spreads plaintext influence across ciphertext — ShiftRows, MixColumns

### S-box

Substitution box — lookup table used in the SubBytes step of AES rounds

### IV (Init. Vector)

Random value used to start CBC; prevents identical blocks producing identical output

### DES

56-bit key standard — deprecated, broken by brute force in 1998

### AES

Rijndael algorithm — 128/192/256-bit keys, global standard since 2001

### Modular Arithmetic

"Clock arithmetic" — numbers wrap at a fixed modulus; core math for cryptography

# Benefits vs. Limitations

## BENEFITS

**Fast**

Efficient for bulk encryption of large datasets

**Simple**

One key for both encryption and decryption

**Secure (AES)**

AES-256 unbroken after 25+ years

**HW Accelerated**

AES-NI CPU instructions accelerate AES

**Widely used**

TLS, disk encryption, Bitcoin wallets, VPNs

## LIMITATIONS

**Key distribution**

Securely sharing the key beforehand is hard

**Scalability**

N parties need N(N-1)/2 unique keys

**No identity proof**

Cannot prove WHO sent a message

**DES weakness**

56-bit key easily brute-forced today

**ECB is insecure**

Identical plaintext blocks reveal patterns

# Key Takeaways

Chapter 3 — Symmetric Cryptography

1. Cryptography provides 5 services: confidentiality, integrity, authentication, non-repudiation, and accountability
2. Symmetric cryptography uses ONE shared key for both encryption and decryption
3. Stream ciphers encrypt bit-by-bit; block ciphers encrypt fixed-size blocks
4. Feistel ciphers achieve security through confusion (S-boxes) and diffusion (permutation)
5. ECB mode is insecure — always prefer CBC or CTR mode in real-world systems
6. DES (56-bit) is broken; 3DES is transitional; AES (128/192/256-bit) is the current standard
7. AES uses 4 per-round operations: AddRoundKey, SubBytes, ShiftRows, MixColumns
8. Bitcoin wallets encrypt local data using AES-256 in CBC mode
9. Symmetric crypto is fast but suffers from the key distribution problem
10. Public key cryptography (Chapter 4) solves the key distribution problem

# Discussion & Q&A

**Q1** Why did DES fail? What does this teach us about key sizes and the pace of technological progress?

**Q2** Blockchain is a public, transparent ledger — what role does symmetric encryption play in this ecosystem?

**Q3** Why is ECB mode a security risk? Describe a real scenario where ECB fails (e.g., encrypted bitmap images).

**Q4** What is the key distribution problem in symmetric cryptography, and why does it motivate asymmetric crypto?

**Q5** Bitcoin uses AES-256-CBC for wallet encryption. What security risks arise if ECB mode were used instead?

Next: Chapter 4 — Public Key Cryptography (RSA, ECC, Digital Signatures)